

- 15** A sphere of mass 20 g undergoes simple harmonic motion with a period of 9.0 s. The speed of the sphere 3.0 s after starting from the equilibrium is 4.0 m s^{-1} .

What is the kinetic energy of the sphere when it next passes the equilibrium position?

A 0 J

B 0.16 J

C 0.21 J

D 0.64 J